Aerodynamics of a Formula 1 car

How does the angle of the front wing affect the product of the cross sectional area and the drag coefficient of a model of a Formula 1 2024 car?

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1. Research design

1.1 Introduction

In a sport like Formula 1, where the average race is won by a difference of 8.8 seconds (MotorSports Etc., n.d.), gaining a hundredth of a second in each corner is the difference between winning millions of dollars or losing everything. The way to gain this time is by optimizing every part of the car to be able to compete against the other teams.

An F1 car has more than 14.500 individual components (Renault F1 Team, n.d.), is developed by a team of more than 1000 engineers (Youson, n.d.) and costs roughly $16 million (beIN SPORTS, 2025), therefore, they are complexly engineered machines. Out of all the components, the study will focus solely on the front-end of the vehicle, particularly, the front wing. The front wing accounts for approximately 10% of the downforce generated by the car (Cuquerella, 2024), although it may not seem significant enough, according to Article 3.9.7 of the FIA Technical Regulations, "[a part of the front wing] may be adjustable for the sole purpose of trimming front wing aerodynamic load" (FIA, 2024, p. 28), therefore, the ease with which the mechanics can adjust the wing to shift the aerobalance of the car makes the front wings' impact incredibly significant. For further reference, "aerobalance" (short for aerodynamic balance) refers to how the total aerodynamic downforce of a car is distributed between the front and the rear axles, varying the Centre of Pressure and altering the behaviour of the car when turning. (Rodríguez, 2015)

The shift in aerobalance is achieved mainly by the variation of the shape and inclination of the front wing (Chudi et al., 2025, p. 01), in a race weekend, the shape of the front wing can not be altered, only the angle at which it is placed can be varied with the use of a manual windscrew mechanism.

Therefore, having seen the importance of the angle of attack of the front wing in the general behaviour of a Formula 1 car, and its flexibility in allowing the angle to be varied, makes the front wing an exceptionally versatile tool for aerodynamic tuning. For this reason, it is relevant to quantify how the Drag Force (F_D) the changes as a function of the angle of attack. Note that selecting drag force as the focus of the study is arbitrary; a natural extension to this IA would be to examine how variations in the front wing's angle of attack influence downforce (negative lift).

The drag force (F_D) is determined by four variables, two of which are influenced by the shape of the car and are therefore affected by the angle of attack of the front wing. These two variables are A (cross-sectional area) and C_D (drag coefficient) (NASA Glenn Research Center, n.d.). Because both terms are dependent on the wing's angle, differentiating them individually is not feasible. Consequently, this study focuses on establishing relationship between the angle of attack and the product $A \cdot C_D$.

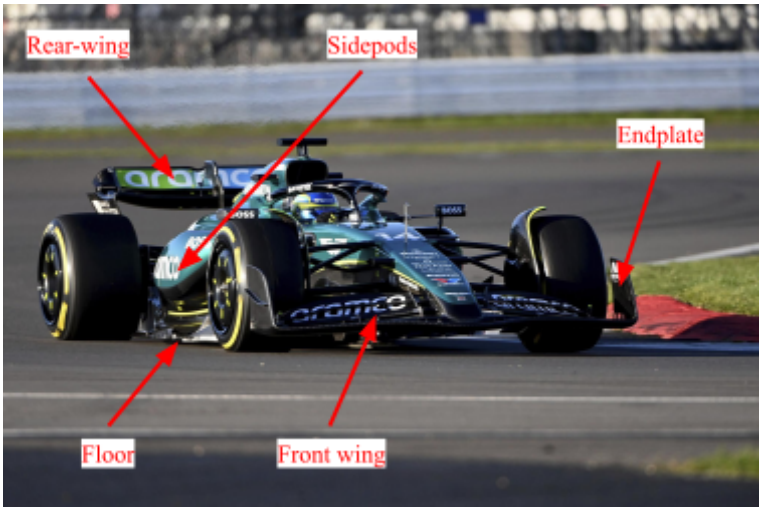


Figure 1) 2024 Aston Martin car with focus on complex aerodynamic shapes (Marca.com)

For these reasons, it is of interest to quantify how variations in the front wing's angle of attack impacts the cross sectional area and the drag coefficient of a Formula 1 car, in this study, specifically the 2024 model, selected due to accessibility. Due to the car's complex aerodynamic shapes (See Figure 1), calculating A and C_D is challenging without relying on approximations or experimental techniques. Therefore, this paper attempts to provide an alternative approach to rapidly calculate the the combined effect of the front wing's inclination on A and C_D, establishing a predictive relationship between the angle of inclination and the resultant $A \cdot C_D$.

The study has been conducted using a LEGO replica of the McLaren model. This model varies in size, but not in shape, as LEGO defines its LEGO® Fórmula 1® products as "realistic and highly detailed replicas", therefore, the results of this study can be extrapolated to the 2023-24 Formula 1 cars (See the comparison between Figures 2a) and 2b).



Figure 2a)



Figure 2b)

Comparison between **Figure 2a)** that shows the 2023-24 model of the McLaren (McL38) and the LEGO replica **(Figure 2b)** (F1Store, n.d.) (LEGO, n.d.)

1.2 Research Question

How does the angle of the front wing affect the product of the cross sectional area and the drag coefficient of the Formula 1 2024 car?

1.3 Background Information

Drag Force (F_D) is a vector quantity that acts in a direction that is opposite to the motion of the body, in this case, the car. (NASA, 2022) However, in this experiment, as the car has no engine, F_D causes the car to go backwards. The equation that allows the calculation of Drag Force is the following:

$$F_D = \frac{1}{2} \cdot \rho \cdot v^2 \cdot C_d \cdot A \text{ [Newtons] (1)}$$

Where:

F_d — the drag force [N]

ρ — the density of the fluid [$kg \cdot m^{-3}$]

v — the speed of the object relative to the fluid [$m \cdot s^{-1}$]

C_d — the drag coefficient [*unitless*]

A — the cross sectional area of the object [m^2]

In order to calculate the product of C_d and A we will need to employ Newton's Second Law of Motion:

$$\Sigma F = m \cdot a \text{ [Newtons] (2)}$$

Where:

ΣF — sum of forces $[N]$

m — mass $[kg]$

a — acceleration $[m \cdot s^{-2}]$

In the case of this experiment, the forces acting on the car are the ones seen in Figure 3), a free body diagram of the F1 model.



Figure 3) Free body diagram of the Formula 1 car in this experiment (F1 Authentics, n.d.)

Note that as the motion of the car is to the right, we will consider forces to that direction to be positive and forces to the left to be negative.

As we can see from Figure 3) the force that opposes motion given by the Drag Force is the Friction force (F_f), which can be calculated using:

$$F_f = \mu \cdot F_N \text{ [Newtons]} \quad (3)$$

Where:

F_f — Maximum Friction Force $[N]$

μ — Silestone's Static Coefficient of Friction [*unitless*]

F_N — Normal Force $[N]$ – and can be expressed as $m \cdot g$

Therefore, using equations (1) (2) and (3), the following equation can be set up in order to isolate the term $C_d \cdot A$

$$F_D + F_f = m \cdot a$$

$$\frac{1}{2} \cdot \rho \cdot v^2 \cdot C_d \cdot A + (-\mu \cdot m \cdot g) = m \cdot a$$

$$C_d \cdot A = \frac{2m(a + \mu \cdot g)}{\rho \cdot v^2} \; [m^2] \; (4)$$

To determine the acceleration of the car for each angle of attack tested, the equation for position will be used.

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2 \; [m(s)] \; (5)$$

Where:

$x(t)$ — Position of the car as a function of time $[m]$

x_0 — Initial position of the car $[m]$

v_0 — Initial velocity of the car $[m \cdot s^{-1}]$

t — time taken for the car to go from x_0 to x $[seconds]$

a — linear acceleration of the car $[m \cdot s^{-2}]$

To get the instantaneous velocity of the car at $x = 0,650m$, equation (6) will be employed.

$$v_{car} = at \ [m \cdot s^{-1}] \ (6)$$

And finally, the equation used to get the relative velocity is as follows:

$$v_{relative} = v_{air} - v_{car} \ [m \cdot s^{-1}] \ (7)$$

The angles tested in this experiment will be 20º, 25º, 30º and 35º. These values were selected because preliminary trials showed that angles below 20º produced very similar results, making any differences negligible due to experimental uncertainties. Similarly, angles greater than 30-40º cause a phenomenon called "flow separation" (Buscariolo et al. 2020), which generates wake effects that reduce downforce experienced by trailing cars, which makes these angles not used in F1.

1.4 Hypothesis

It is hypothesized that the collected data and the subsequent analysis will yield a non-linear positive correlation between the front wing angles and the consequent $C_d \cdot A$ value. This would be a feasible hypothesis because as the front angle increases, the air will be impacting the wing at a greater angle, increasing the drag force (including the $C_d \cdot A$ term) done on the car (to push it backwards). This relationship is predicted to be non-linear, as theoretically, the higher the angle of attack, the faster the car moves at $x = 0.65m$ (as the acceleration is bigger), and this velocity will be squared in the $C_d \cdot A$ formula, which will make the angle of attack proportional to the square of $C_d \cdot A$. According to a CFD study performed by Szudarek, M., & Piechna, J., the positive trend the hypothesis suggests stands, as "CFD results show

that as the front wing angle of attack increases, the aerodynamic drag coefficient increases significantly, demonstrating a clear relationship between wing AOA and drag forces acting on the vehicle." (Szudarek, 2021)

1.5 Variables

Manipulated and measured variables			
Variables		**Explanation**	**Apparatus**
Independent Variable	Angle of attack of the Front Wing	20º, 25º, 30º and 35º Varied by the researcher by adjusting the front wing to different angles	Protractor ±1º (The reading error is the smallest division the physical instrument may read)
Dependent Variable	Time taken to travel from the start and end of the wind tunnel	Measured by recording the end of the tunnel and using a video analysis software to annotate the data	Video analysis software "Tracker" ±0.017 seconds (length of a frame)

Table 1: Manipulated and measured variables

Controlled Variables		
Variables	**Explanation**	**Apparatus**
Wind Speed	The wind speed was measured to be 26,7m/s. As the wind tunnel is 0.65m long, we will assume that this velocity is constant all throughout the tunnel.	This velocity has been calculated using the anemometer "*Thies First Class Advanced II*", which has an uncertainty of 1% (Ammonit Wind & Weather, n.d.) ±1% (26,7 ± 0,267) m/s
Position of the Car	The car was placed right at the beginning of the 0.65m long wind tunnel. With the front wing being at position 0. The hair dryer was placed at a distance of 10 cm from the car.	Measuring tape ± 0,001 m

Table 2: Controlled variables

Physical Constants Used	
Constant	**Explanation**
Gravity (g)	The value of gravity used is 9,81, as the experiment was conducted in the coastal city of Barcelona. (NASA Glenn Research Center, n.d.)
Air Pressure (ρ)	The value of the air pressure used is 1,2250 $kg \cdot m^{-3}$ (Skybrary Aviation Safety, n.d.) *Note: As the experiment was carried out in a coastal city (Barcelona) at average temperatures, it is feasible to accept the International Standard Atmosphere (ISA) value as a correct approximation.*
Static Friction Coefficient (μ)	The value of the static friction coefficient of Silestone used is 1,02. (Cosentino, n.d.)

Table 3: Physical Constants Used

In the calculations presented in the Analysis section, uncertainties associated with the constants above will be neglected. These values are well-established physical constants, and any variation is negligible compared to the uncertainty of the measurements taken.

1.6 Materials

1. 2024 LEGO McLaren Model of a Formula 1 car with an adjustable front wing (LEGO, n.d.)

Size (length, width and height): 19,3 cm x 7,5 cm x 4,4 cm

2. Parlux Hair dryer 2000W: Wind speed calculated using "*Thies First Class Advanced II*" (Used the cold air mode)

3. Cylinder shaped wind tunnel (⌀ 16cm) made with cardboard (65cm long) (See Figure 4)

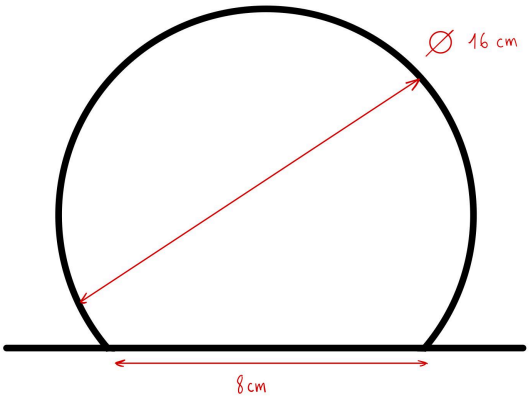


Figure 4) Sketch of the cross-section view of the wind tunnel. Self-drawn using Goodnotes 5. Diagram not to scale.

4. Adhesive tape

5. Measuring tape

6. Camera iPhone 13 (31 FPS, 1080p)

7. Tripod

8. Video analysis software "Tracker"

9. Laptop

1.7 Method

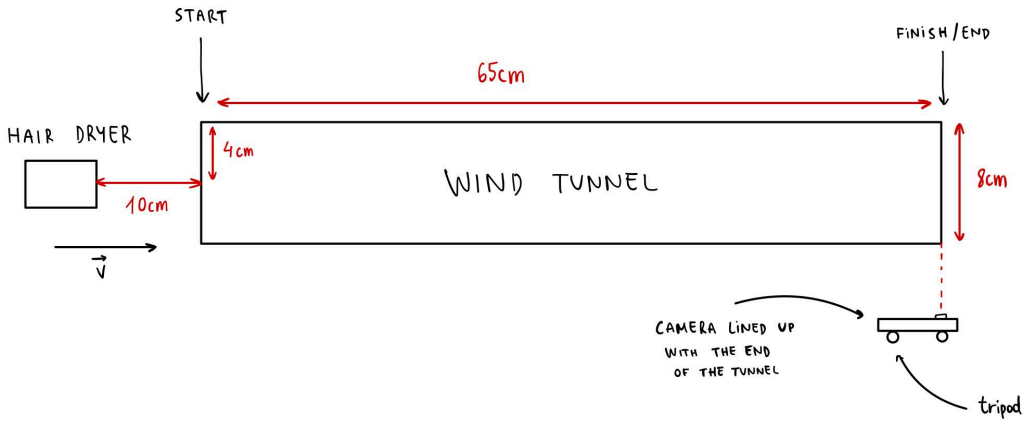


Figure 5) First Experimental Setting. Self-drawn using Goodnotes 5. Diagram not to scale.

1. Clear a smooth and flat surface of 50cm x 100cm.

2. Using adhesive tape, place and fix the wind tunnel within the cleared surface. Note that a 10 cm margin has to be left at the start of the wind tunnel.

3. Fix the hair dryer at a distance of 10 cm from the start of the wind tunnel with adhesive tape. Note that the centre of the hair dryer has to lie in the middle (4 cm) of the wind tunnel's opening.

4. Place the tripod and the camera to the end of the wind tunnel (See Figure 5), so that the side-view of the car is observed as it crosses the end of the tunnel, capturing both sound and video.

5. Adjust the front wing of the car to the desired position (20º, 25º, 30º or 35º) using a protractor.

Note: the "angle of the front wing" refers to the included angle between the front wing and the x-axis (See Figure 6)



Figure 6) Included angle of the front wing (compared with the $x - axis$) Self-drawn using Goodnotes 5. Note that the angle will be considered in absolute value.

6. Place the car at the start of the wind tunnel (See Figure 5) facing the hair dryer, so that when the dryer is turned on, the car travels backwards on a straight line.

7. Start the recording and turn on the hair dryer (mode: MAX / cold)

8. Wait for the car to completely cross the end of the tunnel (See Figure 5)

9. Stop the recording

10. Repeat steps 6-9 five more times

11. Change the angle of attack and repeat steps 6-10 for all the angles

12. Upload one video to the "Tracker" app, mark as $t = 0$ when the highest sound spike occurs (this is when the hair dryer starts) and find the frame when the first part of the car (the backwing) is shown in the image. If the ending between two frames is unclear, take the average time.

13. Note the time and plot it in an excel file.

14. Repeat steps 12 and 13 for all the videos.

Gathering data to do calculations:

15. Position the anemometer at 8cm from the hair dryer (we will assume that the speed of air is constant throughout the wind tunnel).

16. Pointing directly at the anemometer, turn on mode: MAX, cold.

17. Note the reading for further reference.

1.8 Preliminary Trials

While preparing the experiment, preliminary trials were conducted to identify some challenges and fix the conditions to improve the experimental conditions. The main challenge solved was that the wind tunnel's shape was affected by the wind. Two pencils were fixed on both sides of the wind tunnel to make the structure more stable.

1.9 Risk Assessment

Risk Assessment	
Ethical Considerations	No human or animal resources were employed during this study.
Safety Considerations	As a hair dryer was used, it was kept clear of any source of liquid.
Environmental Considerations	The majority of resources used to build the experimental setup (cardboard, LEGO...) were stored and are available to reuse.

Table 4: Risk Assessment

2. Analysis

The data collected for each angle of attack consists of the time taken by the car to move from position $x = 0$ to $x = 0,650\ m$. By using Equation (5), the calculation for acceleration is done for each one of the angles of attack. Hence, substituting the result for a into equation (6) will yield the instantaneous speed of the car at $x = 0,650\ m$. Allowing for the calculation of its relative speed with respect to air, and finally including these values into Equation (4) to find the final $C_d \cdot A$ value.

The data resulting from the experiment is as follows:

QUANTITATIVE DATA (in seconds)				
	20º	**25º**	**30º**	**35º**
Trial 1	1,518	1,401	1,401	1,251
Trial 2	1,435	1,468	1,335	1,351
Trial 3	1,468	1,468	1,468	1,335
Trial 4	1,468	1,418	1,401	1,351
Trial 5	1,435	1,418	1,368	1,368
AVERAGE	**1,465**	**1,435**	**1,395**	**1,331**

Table 5: Quantitative Data (in seconds)

See the following example for Trial 1 of the 35º front wing, which yielded a time of $1,251\ s$. (See Table 5)

Firstly, acceleration is calculated using Equation (5):

$$x\,(t)\ =\ x_0\ +\ v_0 t\ +\ \tfrac{1}{2}at^2$$

$$0,650\ =\ 0\ +\ 0t\ +\ \tfrac{1}{2}a\cdot 1,251^2$$

$$a\ =\ 0,8307\ m\cdot s^{-2}$$

And its relative uncertainty:

$$\tfrac{\Delta a}{a} = \tfrac{\Delta x(t)}{x(t)}\ +\ 2\cdot\tfrac{\Delta t}{t}$$

$$\tfrac{\Delta a}{a} = \tfrac{0,001}{0,65}\ +\ 2\cdot\tfrac{0,017}{1,251}$$

$$\tfrac{\Delta a}{a} =\ 1,54\cdot 10^{-3}\ +\ 2\cdot 0,0136\ =\ 0,0287$$

Secondly, using Equations (6) and (7) the relative speed between the airflow and the object can be calculated:

$$v_{car}\ =\ at$$

$$v_{car} =\ 0,8307\cdot 1,251\ =\ 1,039\ m\cdot s^{-1}$$

$$v_{relative} =\ v_{air} - v_{car} =\ 26,7\ -\ 1,039\ =\ 25,66\ m\cdot s^{-1}$$

Along with its absolute uncertainty:

$$\Delta v_{car} = \left(\frac{\Delta a}{a} + \frac{\Delta t}{t}\right) \cdot v_{car}$$

$$\Delta v_{car} = (0,0287 + 0,0136) \cdot 1,039$$

$$\Delta v_{car} = 0,0439\ [m \cdot s^{-1}]$$

$$\Delta v_{relative} = \Delta v_{air} + \Delta v_{car}$$

$$\Delta v_{relative} = 0,267 + 0,0439$$

$$\Delta v_{relative} = 0,311\ [m \cdot s^{-1}]$$

Note: All future references to v refer to $v_{relative}$, unless stated explicitly.

Substituting $a,\ v_{relative}$ and the boundary conditions (See Table 3) into Equation (4) yields the value for $C_d \cdot A$.

$$C_d \cdot A = \frac{2m(a + \mu \cdot g)}{\rho \cdot v^2}$$

$$C_d \cdot A = \frac{2 \cdot 0,1154\ (0,83 + 1,02 \cdot 9,81)}{1,2250 \cdot 25,66^2} = 3,10 \cdot 10^{-3}\ m^2$$

The calculation of the relative and absolute uncertainty for $C_d \cdot A$ is:

$$\frac{\Delta(C_d \cdot A)}{C_d \cdot A} = \frac{\Delta m}{m} + \frac{\Delta a}{a} + 2 \cdot \frac{\Delta v}{v}$$

$$\frac{\Delta(C_d \cdot A)}{C_d \cdot A} = 8,6655 \cdot 10^{-4} + 0,02871 + 2 \cdot 0,01211$$

$$\frac{\Delta(C_d \cdot A)}{C_d \cdot A} = 0,0538$$

$$\Delta(C_d \cdot A) = 0,0538 \cdot 3,101 \cdot 10^{-3}$$

$$\Delta(C_d \cdot A) = 1,6 \cdot 10^{-4}\,[m^2]$$

Therefore, the final calculated value of $C_d \cdot A$ for 35º with its absolute uncertainty is:

$$C_d \cdot A = (3,1 \pm 0,2) \cdot 10^{-3}\, m^2$$

Note: As the relative uncertainty is relatively large (5,38%), and it is bigger than 2%, the uncertainty is reported to one significant figure, in accordance with standard uncertainty conventions.

Repeating this same procedure for all trials and each angle of attack tested yields the following table (Table 7):

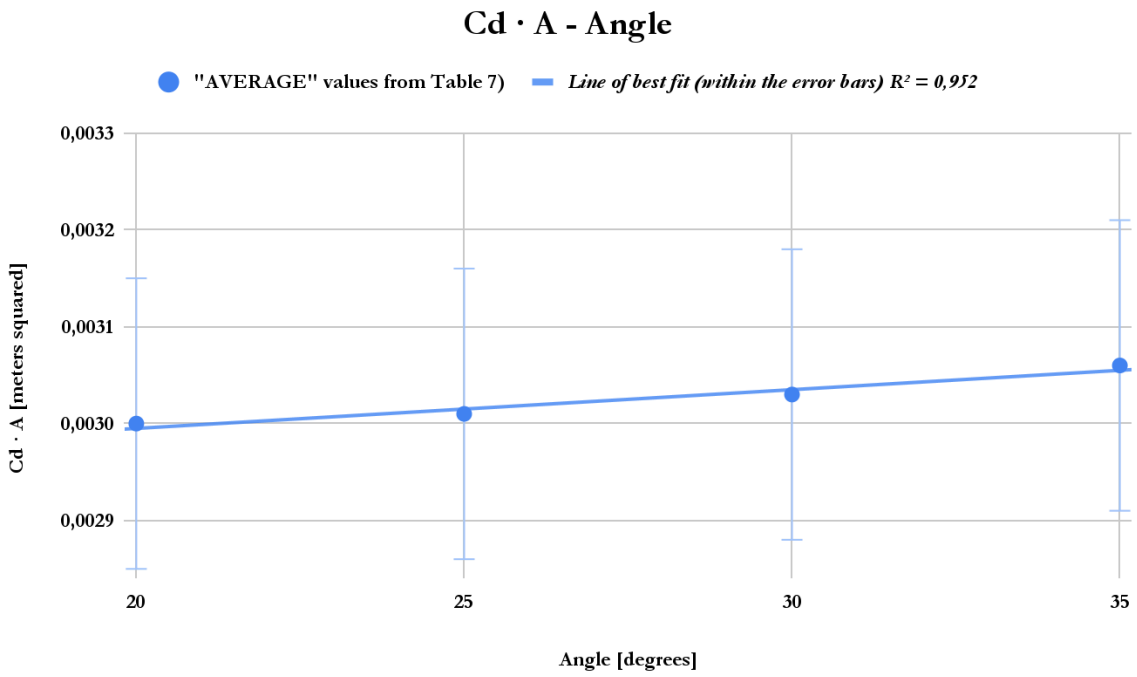
Results Table $[m^2]$

$C_d \cdot A$	**20º**	**25º**	**30º**	**35º**
Trial 1	$(2.98 \pm 0.1) \cdot 10^{-3}$	$(3.03 \pm 0.2) \cdot 10^{-3}$	$(3.03 \pm 0.2) \cdot 10^{-3}$	$(3.10 \pm 0.2) \cdot 10^{-3}$
Trial 2	$(3.01 \pm 0.2) \cdot 10^{-3}$	$(3.00 \pm 0.2) \cdot 10^{-3}$	$(3.06 \pm 0.2) \cdot 10^{-3}$	$(3.05 \pm 0.2) \cdot 10^{-3}$
Trial 3	$(3.00 \pm 0.2) \cdot 10^{-3}$	$(3.00 \pm 0.2) \cdot 10^{-3}$	$(3.00 \pm 0.2) \cdot 10^{-3}$	$(3.06 \pm 0.2) \cdot 10^{-3}$
Trial 4	$(3.00 \pm 0.2) \cdot 10^{-3}$	$(3.02 \pm 0.2) \cdot 10^{-3}$	$(3.03 \pm 0.2) \cdot 10^{-3}$	$(3.05 \pm 0.2) \cdot 10^{-3}$
Trial 5	$(3.01 \pm 0.2) \cdot 10^{-3}$	$(3.02 \pm 0.2) \cdot 10^{-3}$	$(3.04 \pm 0.2) \cdot 10^{-3}$	$(3.04 \pm 0.2) \cdot 10^{-3}$
AVERAGE	$(3.00 \pm 0.2) \cdot 10^{-3}$	$(3.01 \pm 0.2) \cdot 10^{-3}$	$(3.03 \pm 0.2) \cdot 10^{-3}$	$(3.06 \pm 0.2) \cdot 10^{-3}$

Table 7) $C_d \cdot A$ values for all trials and average between trials

The averages are displayed in the last row of Table 7)

Using the "AVERAGE" values from Table 7), an Angle - $C_D \cdot A$ graph (Graph 1) is created. It includes error bars with the average absolute uncertainty and the line of best fit, showing a linear positive correlation of $R^2 = 0.952$.



Graph 1) Average values for $C_d \cdot A$ for each angle plotted on a graph

3. Conclusions

The result of this Internal Assessment is found in Graph 1). It shows a linear relationship between the angle of attack and the $C_d \cdot A$ value. Even though the values themselves would fit a non-linear curve — as it was hypothesized — the uncertainties of around 5% make this result inconclusive. Therefore, the angle of attack of the front wing affects the cross-section area and the drag coefficient according to the following function:

$$C_d \cdot A\,(x) = 0.000004x + 0.002915$$

Where:

$C_d \cdot A$ — product of the drag coefficient and the cross-section area [m^2]

x — degrees of inclination of the front wing [degrees]

Using this equation, the calculation of the $C_d \cdot A$ values corresponding to each angle of attack of the LEGO model is possible. The trend that this equation yields is consistent with the paper cited in the hypothesis (Szudarek, 2021), as it is a positive linear relationship between both variables.

4. Evaluation

Uncertainties

The average uncertainty of the experiment was 5.01%. With the maximum uncertainty being 5.18% and the minimum being 4.88%. This shows consistency in the collection of data and its consequent processing. Five trials were conducted for each angle of attack, reducing the possibility of statistical and random errors.

Weaknesses and Limitations

RANDOM ERRORS		
LIMITATION	**REASON**	**IMPROVEMENT**
Airflow speed fluctuations	The hair dryer has small and random variations in output velocity in each run.	Perform 10 more trials, exclude outliers and take an average to find a truer value.
Video frame-rate timing uncertainty	Whether the crossing of the rear wing occurs near or the end of a frame is random.	Use a higher-quality slow-motion camera to reduce the uncertainty.
Minor variations in initial car placement / alignment	Millimetric differences between trials.	Use position sensors to verify that the initial position of the car is consistent and accurate for all trials.

Table 8) Random Errors

SYSTEMATIC ERRORS		
LIMITATION	**REASON**	**IMPROVEMENT**
Camera's FPS	My iPhone 13's camera records at 31 FPS, therefore, due to the short difference between angles of attack, there is raw data that coincides.	Use a high-quality slow motion camera (100 FPS) to have more frames to choose from, and therefore find a more exact timing. Doing so would reduce the uncertainty from ($\pm$ 0.017 s) to ($\pm$ 0.005 s)
Hair dryer's airflow	The Drag Force equation assumes steady and laminar airflow, for simplicity, we have assumed, but can not confirm that the hair dryer allows for this phenomenon.	Redo the experiment in a special chamber that produces laminar flow.
Wind-tunnel non-steadiness	Even though the pencils helped the wind tunnel remain stable, the fast airflow still made the structure not steady, which may result in a more disturbed airflow.	Substitute cardboard with other materials such as Polycarbonate (PC) to reduce wobbling.

LEGO Model	Even though the model is based off the 2024 model, LEGO models don't allow for smooth and continuous aerodynamic components.	Buy a more detailed replica that mimics the car's components, but at the same, allows the front angle to be modified, which is the reason why it was not used in this experiment.
Friction Coefficient	Due to not finding a rotation coefficient for Silestone and rubber, and as the tyre did not slip, but rotate, a static situation has been assumed in order to be able to perform the friction force calculations.	A parallel experiment could be conducted in order to find the rotational friction coefficient.

Table 9) Systematic Errors

LIMITATIONS		
LIMITATION	**REASON**	**IMPROVEMENT**
Scaling / Reynolds number mismatch	Due to the models' reduced size, $C_d \cdot A$ values cannot be extrapolated to full-scale cars	Apply correction factors from CFD studies and match the Reynolds number wherever possible.
Simplified wind tunnel geometry	Creates a confinement of airflow due to the walls. The pressure distribution and drag coefficients differ from those in free-air conditions. (Tabatabaei et al., 2021)	Use larger wind tunnel sections to minimize blockage effects. (Tabatabaei et al., 2021)
Focus on Cd · A	Due to both terms being incognitas and depending on the wing's inclination, the investigation had to be carried out by merging both terms.	Apply multi-variable regression to isolate the effects of each term. (Great Learning Editorial Team, 2024)

Table 10) Limitations

Extensions

This Internal Assessment could be extended by using a bigger scale model in order to minimize the effect of the uncertainty on the final result and find a clearer non-linear correlation. Another extension would be analyzing how different components of the car affect the cross sectional area of the car and the drag coefficient, for instance, designing different side-pods, rear-wings or even floor. Moreover, this specific exploration could be also expanded by not only varying the angle of attack, but also changing the shape and size of the front-wing.

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